SMART INDIA HACKATHON 2025

- Problem Statement ID – SIH25262
- Problem Statement Title- AI based real-time crop image analytics for crop insurance-PMFBY
- Theme- Agriculture, FoodTech & Rural Development
- PS Category- Software
- Team ID- 93163
- Team Name- Imaginary_coders

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IDEA TITLE

Idea & Solution — Detailed Description

Farmers capture periodic field photos via the CROPIC app — AI models analyze images to detect crop stage, stress, pest/disease, and yield potential. Integrated APIs pull district/state circulars, weather, and market prices to personalize insights and alerts.

3.How it functions in practice.

- Farmers track insurance claim status in-app and receive payments directly in a Digital PMFBY Wallet, eliminating middlemen. AR-guided capture, microlearning videos, and helpline chat reduce anxiety and upload errors.
- ***Grievance Redressal A dedicated system allows farmers to file complaints for unpaid or underpaid insurance claims by uploading supporting proof verified by local authorities.***
- Every farm plot becomes a verified digital node. Farmers upload geotagged crop photos; AI cross-validates them with satellite & weather data. The grid auto-generates crop condition heatmaps for each district — used for instant insurance verification and subsidy targeting.
- ***Satellite Monitoring Farmers can access satellite imagery via the app to proactively monitor their fields for emerging calamities, such as floods.***

2. How it addresses the problem

- **Real-Time Crop Monitoring:** Provides continuous farm-level crop health, stress, pest/disease, and growth stage data instead of fragmented reports, supporting timely interventions and crop insurance decisions.
- **Market & Financial Planning:** Integrates real-time commodity prices and yield forecasts, helping farmers decide when and where to sell crops, improving income and market efficiency.
- **Community Engagement:** Farmers and AgriChampions verify and contribute geotagged crop data, ensuring accuracy, transparency, and trust in the system.
- **Digital Literacy:** AR-guided uploads, microlearning, and multilingual voice assistance reduce confusion and anxiety for low-literacy farmers, ensuring broader adoption.

4. Gamification & Work Management

- Teams & Units: AgriChampion groups/farmer clusters complete daily/weekly objectives (crop uploads, pest reports, stress monitoring).
- Scoring & Leaderboards: Improvement % = (Initial – New Stress Index)/Initial × 100; top performers earn Gold/Silver/Badge points.
- Crop Quests: Tasks like uploading stage-specific crop images 🌅, tagging pests 🐛, and comparing greenness with district average.
- Points & Rewards: AgriPoints redeemable for fertilizer/seed discounts, priority insurance processing, or PMFBY wallet cashback.
- Feedback & Community Validation: AI adapts challenges; verified farmer/AgriChampion reports strengthen trust and data quality.

TECHNICAL APPROACH

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USER INTERACTION

- Farmer App
- AgriChampion App
- Citizen Reporter

BACKEND & ENGAGEMENT

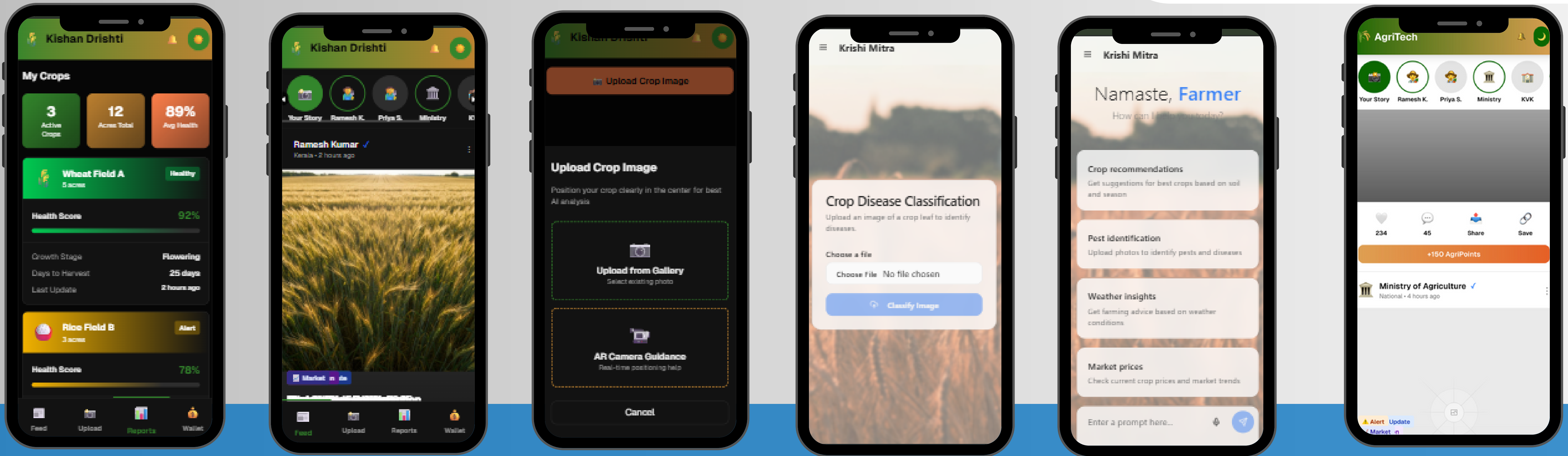
- Central Database
- Auth Service
- Claim Management
- PMFBY Wallet
- Gamification Engine
- Policy Feed API
- Claim Tracker
- Notification Service

CLOUD GATEWAY & AI/ML

- Weather API
- Storage
- API Gateway
- Queue
- Soil & Geo Data
- Satellite Imagery

AI/ML ENGINE

- Crop ID
- Growth Stage
- Image Pre-Processing
- Data Fusion Service
- Pest/Disease
- Yield Prediction



GOV DASHBOARD & FEEDBACK

- Feedback Portal
- AI Feedback Analyzer
- Admin Panel
- Government Dashboard

FEASIBILITY AND VIABILITY

IMAGINARY CODERS

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Potential Challenges & Risks

01. Fragmented Data Flow → Lack of real-time bridge between government and farmers.

02. Fraudulent Image Uploads → Manual verification can't detect manipulation.

03. Administrative Delays → Claims, policy updates, and DBT often bottlenecked by intermediaries.

04. Connectivity Gaps → Rural network instability delays communication.

05. Data Privacy & Compliance → Sensitive user data must follow PMFBY, Aadhaar, and Data Protection guidelines.

1. Technological Feasibility:

- AI engine using CNNs and Vision Transformers analyzes crops offline via Edge AI and auto-syncs when online.

2. Operational Feasibility:

- AgriChampion verification with geo-tagging and AR image capture ensures reliable, standard data from low-end devices.

3. Economic Viability:

- Digital verification reduces agent costs; automated Direct Benefit Transfer (DBT) ensures transparent, fraud-free payouts.

4. Social & Policy Viability:

- Local-language voice interface boosts inclusion; web community page links farmers directly with the government.

5. Implementation Feasibility:

- Open APIs and standardized protocols enable smooth integration with government databases and easy deployment across regions.

1. Cost Optimization:

- AgriChampion verification cuts intermediary costs; cloud automation streamlines claims and reduces admin work.

2. Revenue Generation:

- Partnerships with AgriTech firms create shared income streams, supported by CSR and institutional funding.

3. Scalability:

- Microservice design allows state-wide scaling; open APIs enable smooth future integrations.

4. Government Alignment:

- Integrates with PMFBY, eNAM, and IMD, ensuring policy support and easy adoption via government digital frameworks.

5. Sustainability:

- Gamified, local-language interfaces boost engagement, while predictive analytics enhance transparency and trust among farmers.

Strategies to overcome

01. Digital Inclusion: Voice-assisted interface, regional languages, and offline access ensure no farmer is left out.

02. Policy Integration: Seamless APIs (PMFBY, eNAM, IMD) integration allows Ministry to monitor crop groups and deliver policies directly.

03. Transparency & Trust: Blockchain-inspired records and geo-verification eliminate manipulation and build trust.

04. Data Accuracy: Multi-source data fusion (satellite + IoT + AI) refines stress, yield, and claim predictions.

05. Farmer Engagement: Interactive community page with AgriPoints rewards, feedback, and awareness drives sustained participation.

IMPACT AND BENEFITS

IMAGINARY CODERS

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SIH

Economic & Risk Reduction

→ Faster, Fraud-Free Claim Settlement: Direct Benefit Transfer (DBT) via PMFBY Wallet and Blockchain-Inspired Claim Ledger ensures transparent, traceable, and significantly faster settlement of insurance claims.

Knowledge & Loss Prevention

→ Real-Time, Localized Crop Intelligence: Provides instant, offline-first early warnings for pests, diseases, and weather (via Edge AI) to prevent losses and guide timely action.

Accessibility & Inclusion

REAL TIME DATA

→ Overcomes Digital Divide: Voice-based UI in regional languages and support for SMS/IVR ensures adoption by non-literate and low-connectivity farmers.

For Farmers

- Transparent claim tracking and faster settlements via PMFBY Wallet.
- Early warnings on crop stress, pests, and weather to prevent losses.
- Access to market insights → better price decisions, reduced exploitation.
- Reduced anxiety through guided uploads, local helpline, and micro-learning.

For Market & Economy

- Predictive yield data improves supply-chain planning and reduces wastage.
- Encourages market-responsive cropping and boosts farmer income.
- Creates a new ecosystem of AgriChamps, startups, and research collaborations for long-term growth.

For Government

- Real-time, map-based dashboard for crop health, yield forecasts, and subsidy planning.
- Better disaster response and data-driven policymaking.
- Eliminates middlemen and fraud through digital traceability and youth-led field validation.

Governance & Policy

→ Data-Driven Decision Making: Provides a Real-Time, Map-Based Dashboard with accurate crop health, stress predictions, and yield forecasts, enabling superior disaster response and targeted subsidy planning.

Efficiency & Trust

→ Elimination of Middlemen & Fraud: Blockchain-backed claim tracking and the AgriChampion Youth Verification Portal replace traditional agents, drastically reducing administrative overhead and

Verification & Accuracy

→ AI-Verified Insurance Workflow: Fuses ground-level image data with Satellite-AI Fusion (NDVI) to automatically cross-check claims, ensuring high accuracy and preventing false claims.

UNIQUE SELLING POINT

Satellite Verification: Farmers view their land with Live Satellite Overlay (NDVI) to monitor crop health and boundaries. This verifiable visual evidence is linked to digital land records.

Voice-driven Complaint System: Farmers speak their issues (pests, damage, policy confusion) directly into the app. System creates a Geo-tagged digital ticket.

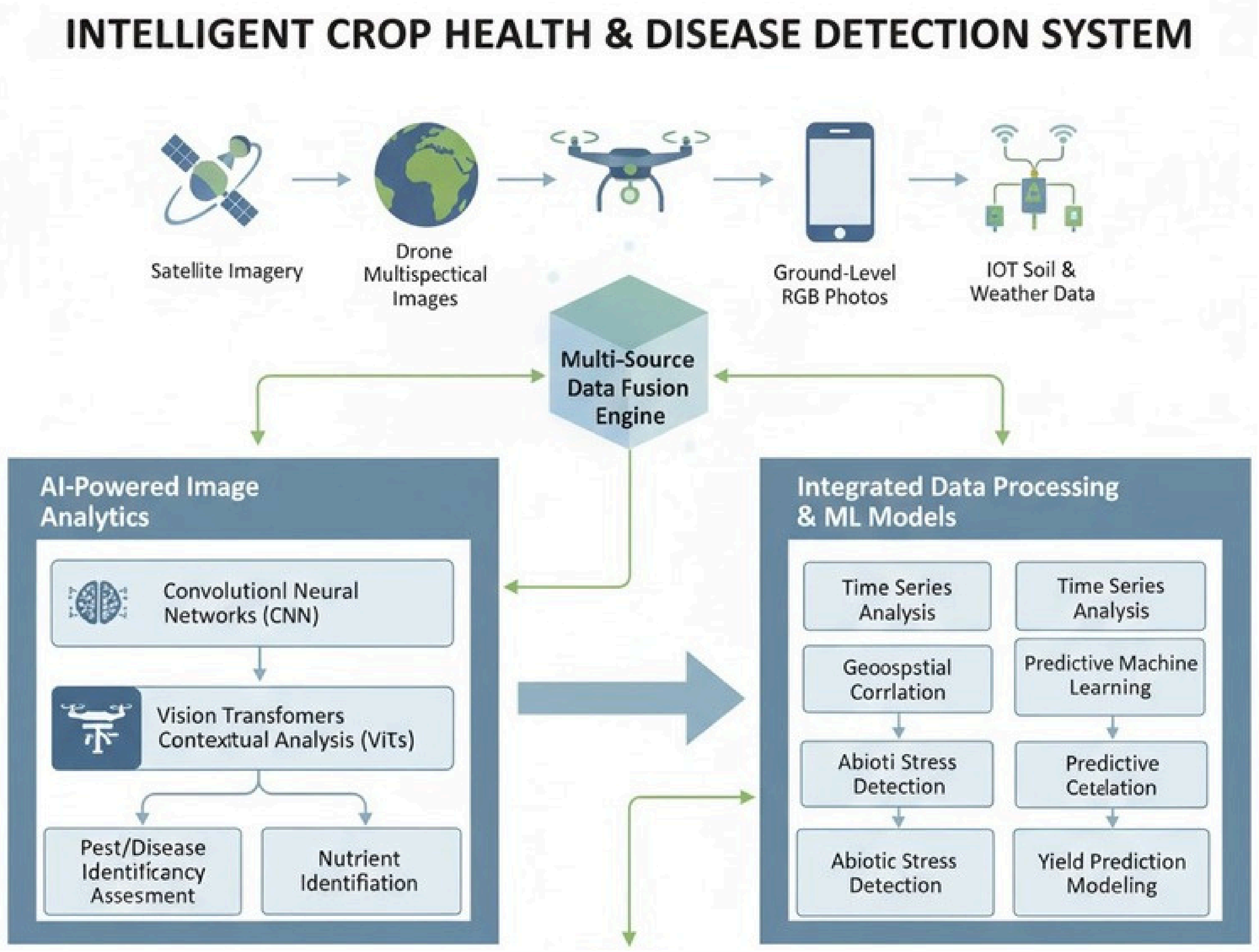
Offline-First, Farmer-First Hybrid System: Unlike cloud-heavy competitors, our system uses Edge AI (TensorFlow Lite) for instant, on-device analysis, making it the most reliable solution in low-connectivity rural India.

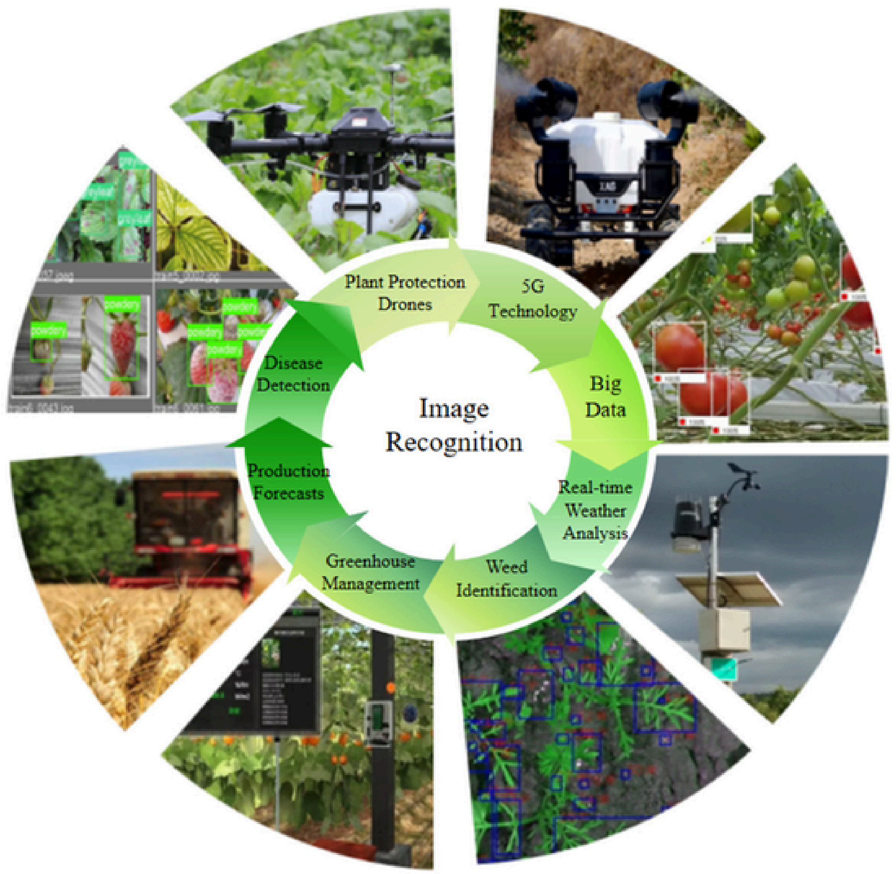
"TrustChain" — Blockchain-Inspired Claim Assurance: Ensures tamper-proof evidence and transparent claim history for both the farmer and the insurer, a critical feature for high-value insurance claims.

Dual-Layer Fraud Detection: Combines AI image authenticity checks (EXIF data, GPS, lighting) with Satellite Data Cross-Verification (NDVI) to create a robust, AI-verifiable insurance claim process.

Highest Adoptability Model: Achieved by combining a Voice/Conversational UI for non-literate users with low infrastructure requirements (basic Android, SMS/IVR fallback) and community adoption through AgriChampions.

RESEARCH AND REFERENCES





Leveraging Multi-Modal Data & Deep Learning for Precision Agriculture

Research Paper Articles-

Image Recognition Technology in Smart Agriculture: https://www.mdpi.com/2227-9717/13/5/1402

Croplife.Com: www.croplife.com/editorial/best-agriculture-appseplergl/kepler.gl - list of the best mobile apps in agriculture since 2010.

Report link-

https://drive.google.com/drive/folders/1hwojBC2dbQx4U-djM89H4t4ZufJwAwGs

Reference github repository-

Core Platforms: AI Model

- **Supports satellite imagary and data fusion:** https://github.com/microsoft/farmvibes-ai
- **Agri-VisionAI:**https://github.com/4bdul4ziz/AgriVision agricultural ML framework.

Visualization And Satellite Analysis:

- https://github.com/umair1221/AI-in-Agriculture - survey & resources for AI in agriculture.
- **Satellite analysis for agriculture.:** https://github.com/sanjith-3/agrisat-ai - **Agrisat-AI**

Github link for AI model:

https://github.com/Sahu-Chitransh/krishi-chat-backend

Team Members:-

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GITHUB Repo Link-

https://github.com/AnuragTiwari1508/Krishi-drishya

Prototype Link-

https://KishanDrishti.vercel.app